Measuring Muscle Fatigue With Electromyography

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Abstract

Muscle fatigue can be difficult to measure due to the wave signal. Due to being generated by an EMG-sensor, it is not always accurate and therefore can be challenging to read. As such, obtaining correct data from the electric signal generated by the muscle is crucial for the determination of muscle fatigue. This project aims to collect as much data as possible and develop a pattern recognition system that can determine if the muscle is exhibiting signs of fatigue. The prototype recorded muscle activity and displayed changes in RMS and MAV, but further data collection and testing are needed to establish a reliable threshold for identifying muscle fatigue.

Introduction

Surface electromyography (sEMG) can be used to evaluate muscle fatigue by measuring the electrical activity of the muscle. However, the signal is affected by factors such as noise, sensor properties and electrode placement. Therefore making reliable signal acquisition a priority before fatigue can be analyzed [1], [2]. Previous research indicates that muscle fatigue can be identified by analyzing features of the sEMG signal. Fuentes del Toro et al. used wavelet filtering and evaluated RMS (Root Mean Square), MAV (Mean Absolute Value), and MNF (Mean Frequency). Showing that RMS and MAV increased while MNF decreased as fatigue developed [3].

This project aims to evaluate an sEMG measurement system that amplifies, filters, and records muscle activity. The recorded data is analyzed over time using RMS, MAV, and MNF to investigate whether a consistent fatigue related pattern can be identified.

Design

This project presents a wearable sEMG measurement system consisting of three electrodes with a signal processing device as shown in figure 2. The electrodes detect the electric signals generated by the muscle, which are then amplified and filtered using an analog circuit powered by two 9V batteries. An ESP32 converts the conditioned analog signal to a digital signal by using its built-in ADC, which is controlled by C++ code. The acquired data can then be transferred via Wi-Fi to a computer, where a python program calculates RMS and MAV and then plots the results for analysis of muscle activation and fatigue[figure 1].

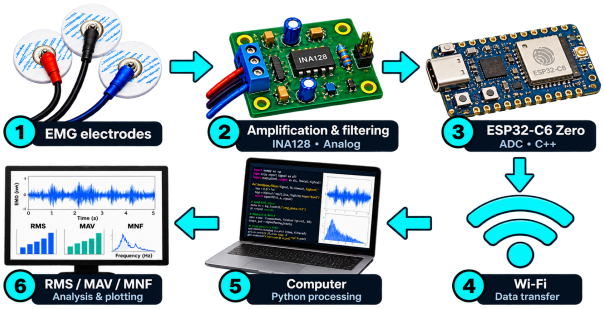


Figure 1: System overview of the signal flow from sEMG electrodes to signal processing, data transfer and result visualization (Image generated by AI).

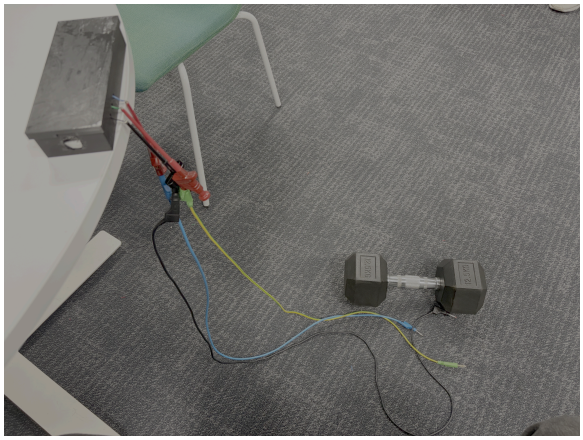


Figure 2: The developed sEMG measurement system, with all electronic components housed inside the black box. The measurement electrodes are connected via the blue and yellow cables, while the reference electrode is connected via the black cable.

Evaluation

The prototype was evaluated during repeated biceps curls using a 10 kg load. A total of 30 recordings were collected from 4 participants. Two measurement electrodes were placed over the biceps, while a reference electrode was placed near the wrist. The recordings were acquired using the ESP32 and transmitted via Wi-Fi to the computer for processing in Python. Figure 3 presents approximately 50 seconds from one of the 30 recordings, showing the processed sEMG signal together with RMS and MAV calculated using a 0.25-second analysis window. The evaluation examined whether the system could record changes in muscle activity and provide amplitude based features for following analysis.

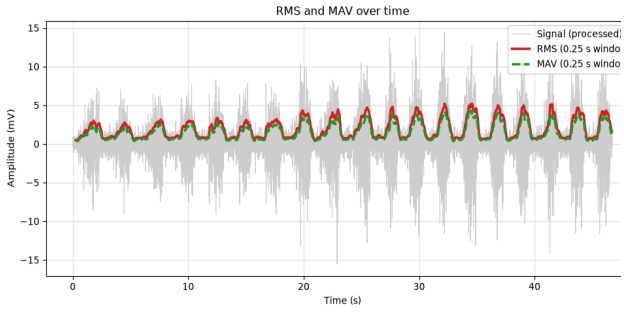


Figure 3: Processed sEMG signal (grey) with RMS (red) and MAV (green dashed), shown over a 50 seconds window. The curves show changes in signal amplitude over time during repeated muscle contraction.

Visual inspection of Figure 3 shows repeated variations in signal amplitude during the exercise. RMS and MAV followed these variations, with higher peaks visible during later parts of the recording compared with the beginning. Similar patterns were observed in the other recordings, although the magnitude of these changes varied. This demonstrates that the recorded signal could be processed and displayed as amplitude based features.

MNF was not included in the final analysis because we could not obtain reliable estimates of the signal's frequency content. We therefore focused on RMS and MAV, which were sufficient for exploring changes in signal amplitude during exercise.

The participants reported feeling increasingly fatigued as the exercise progressed, which was consistent with the observed increases in RMS and MAV. These reports were noted as additional qualitative support for the observed pattern and were not included in any calculations. However, this agreement alone does not validate the system's ability to detect muscle fatigue.

A demonstration of the prototype is available at [Video link]. The demonstration shows the operation of the complete battery powered prototype. The electrodes detected a signal from the muscle, which was amplified and filtered by the analog circuit. The ESP32 successfully transmitted the acquired data via Wi-Fi to the computer application, where RMS and MAV were calculated and visualized. The application also saved the recording and allowed previously saved recording to be accessed.

The prototype recorded muscle activity and displayed changes in RMS and MAV, which can contribute to these assessments of muscle fatigue. Previous research has reported increases in these features during fatiguing exercise [3]. However, substantially more data and testing are needed to establish and validate a reliable threshold for identifying muscle fatigue. Such a threshold has not yet been established in this project.

Conclusion

This project demonstrated that the developed sEMG system could record clear muscle activity signals and calculate RMS and MAV using a Python program. During repeated biceps curls with 10 kg weight, signal amplitude generally increased alongside reported fatigue. Similar patterns were observed across other tests. However, the collected data were insufficient to establish a reliable fatigue threshold. Future work should include more testing with participants and more evaluation of the data.
The device will also be made lighter and more compact. An adjustable elastic band will wrap around the limb and hold the electrodes in place over the muscle being measured. The box containing all electronic components will be attached to the same band, allowing the user to wear the complete system during exercise.

References

[1] M. Cifrek, V. Medved, S. Tonković, and S. Ostojić, "Surface EMG based muscle fatigue evaluation in biomechanics," Clinical Biomechanics, vol. 24, no. 4, pp. 327–340, 2009. Doi: https://pubmed.ncbi.nlm.nih.gov/19285766/

[2] H. J. Hermens, B. Freriks, C. Disselhorst-Klug, and G. Rau, "Development of recommendations for SEMG sensors and sensor placement procedures," Journal of Electromyography and Kinesiology, vol. 10, no. 5, pp. 361–374, 2000. Doi: https://pubmed.ncbi.nlm.nih.gov/11018445/

[3]Fuentes del Toro, S., Santos-Cuadros, S., Olmeda, E., Álvarez-Caldas, C., Díaz, V., & San Román, J. L. (2019). Is the use of a low-cost sEMG sensor valid to measure muscle fatigue? *Sensors, 19*(14), 3204.

Doi: https://doi.org/10.3390/s19143204

GitHub link:
https://github.com/atb-2025/muscle-fatigue